The eREV Powertrain from VIA Motors

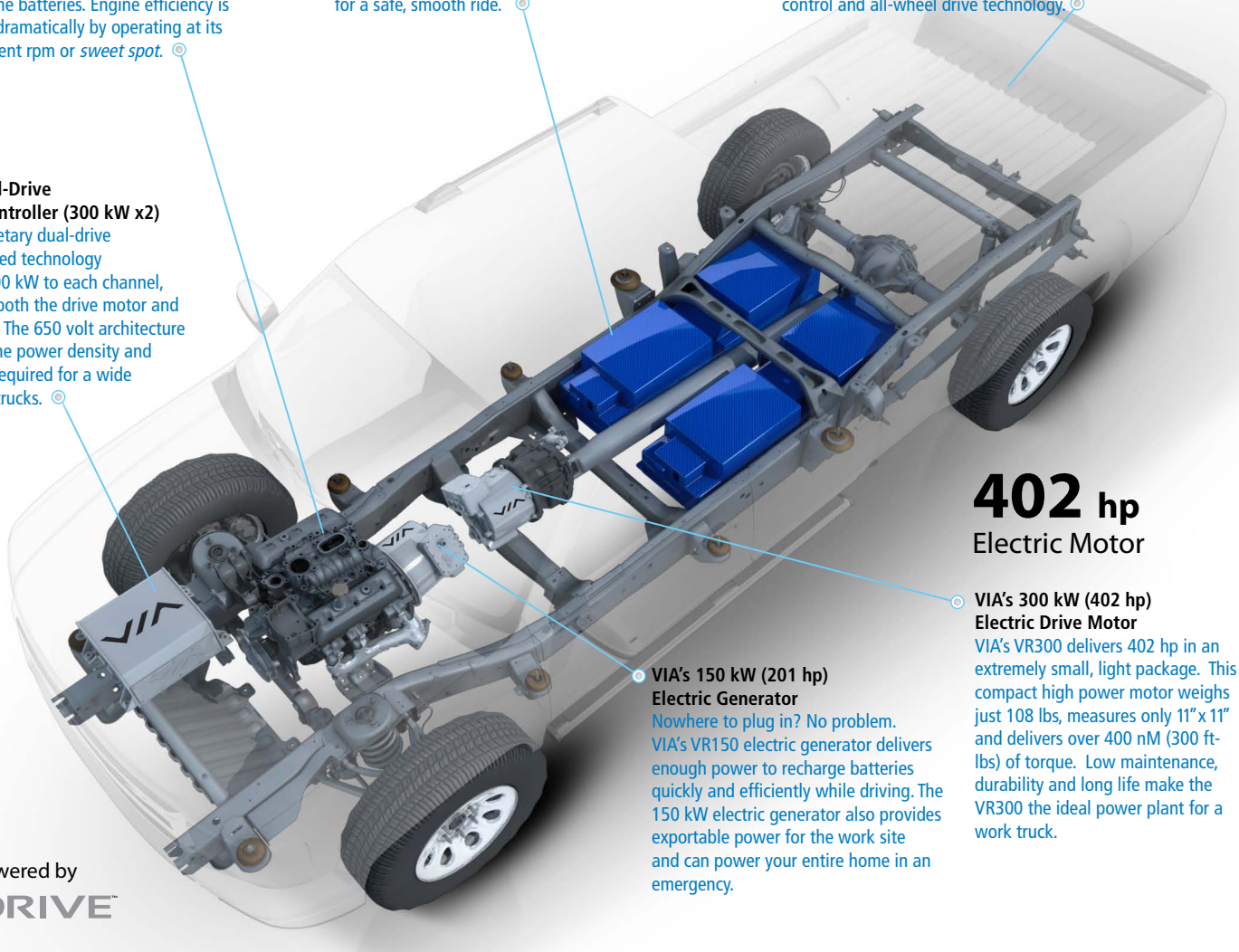
Extended Range Electric Vehicle

Gas/Electric Generator
The economical 4.3L V6 combustion engine is paired with VIA's electric generator to form a gas/electric generator set. The combustion engine is connected directly to the electric generator and is used only when needed to generate electricity to automatically recharge the batteries. Engine efficiency is improved dramatically by operating at its most efficient rpm or *sweet spot*.

Advanced Li-Ion Batteries
The 24.4 kWh liquid cooled Li-ion battery pack delivers up to 40-mile, zero-emission battery range and many hours of quiet power export for the work site. The batteries are located under the bed, safely inside frame rails, giving the truck a low center of gravity for a safe, smooth ride.

Payload and Towing
The VTRUX Extended Cab has a curb weight of 5,500 lbs and a full size bed; delivering a 1,000 lb payload capacity. The 402 hp high-torque electric motor gives ample towing capacity. VTRUX are available in 2WD and 4WD configurations with automatic traction control and all-wheel drive technology.

VIA's Dual-Drive Motor Controller (300 kW x2)
The proprietary dual-drive liquid-cooled technology delivers 300 kW to each channel, powering both the drive motor and generator. The 650 volt architecture provides the power density and economy required for a wide variety of trucks.



402 hp
Electric Motor

VIA's 300 kW (402 hp) Electric Drive Motor
VIA's VR300 delivers 402 hp in an extremely small, light package. This compact high power motor weighs just 108 lbs, measures only 11" x 11" and delivers over 400 nM (300 ft-lbs) of torque. Low maintenance, durability and long life make the VR300 the ideal power plant for a work truck.

VIA's 150 kW (201 hp) Electric Generator
Nowhere to plug in? No problem. VIA's VR150 electric generator delivers enough power to recharge batteries quickly and efficiently while driving. The 150 kW electric generator also provides exportable power for the work site and can power your entire home in an emergency.

Powered by VDRIVE™



Vehicle Configuration
Light Duty Truck 4 x 2 Extended Cab

Feature	Specification
GVWR	7000 lbs
Payload	1000 lbs
Vehicle Dimensions	
Overall Length	230.2 inches
Overall Width	79.9 inches
Overall Height	73.9 inches
Wheelbase	143.3 inches
Suspension	
Front Axle Capacity	3200 lbs
Rear Axle Capacity	3750 lbs
Steering Description	42 Volt Electric Rack Power Steering
Drive Train	4 x 2 2012 Models to include 4 x 4
Seating Capacity	6
Cab Format	2011 – Extended Cab
Traction Motor Description	
Motor Type	VR300 PMSM – Oil Cooled Regen capability on coast and braking
Power (Cont./Peak)	170 kW / 305 kW
Torque	230 Nm / 350 Nm
Base / Max Speed	7,900 rpm / 10,000 rpm
Transmission	One speed step-down gear box
Aux Power Unit	
Gas Engine	GM 4.3L V-6 MFI Gasoline Engine
Generator	VR150 PMSM Oil Cooled
Power (Cont./ Peak)	110 kW / 160 kW
Tires and Wheels	265/70 R17 Steel

Feature	Specification
Brake Description	Hydraulic with vacuum booster includes traction and stability control
Engine Format	eREV
Battery Description	
Capacity	24.4kWh Liquid cooled
Chemistry	LiFePh
Voltage	650 V Nominal
Typical SOC Usage	100 – 20% SOC
Safety Features	CAN 2.0V controlled BMS (cell balancing, health, temp., voltage, etc.), pre-charge circuitry, external manual disconnects, auto power de-rating, internal fuses, diodes, current sensors and relays, etc.
Charger	6.6kW Level I and II Charging SAE J1772 Compliant 4 hr charge time using 240V 15hrs charge time using 120V
Battery Range	30 – 45 miles All Electric Range (AER) depending on driving conditions
Maximum Vehicle Speed	85 mph
Acceleration (0 – 60 mph)	9.7 seconds
HVAC	650V Electric AC Compressor for cabin cooling. Conventional cabin heat from engine coolant loop plus electric heat for defrost
Telemetry Systems	Current vehicles equipped with WiFi for up to 4 users. Other systems can be added at customers' request for data transfer, etc.
Export Power	15kW with four 120V and one 240V outlets in bed of truck Runs off of batteries until battery reaches low SOC; then engine comes on automatically.

VIA Motors. A Better Way to Go.

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